

Test Scenarios

Backend, Frontend and Semantic Testing — Drone Routing Problem

Backend Testing

BT-01	Gurobi Optimization — Valid Request Processing	Pass
Preconditions	The FastAPI backend is running. A valid routing dataset is available.	
Test Steps	1. Select Gurobi as the optimization method in the request payload. 2. Submit the routing problem instance to the /optimize endpoint via POST request. 3. Capture the HTTP response and inspect its structure. 4. Verify that the response contains routes, objective value, and execution statistics.	
Expected Result	The backend returns HTTP 200 with a well-formed JSON body containing generated routes, the objective value, and execution time. The response structure matches the frontend contract.	

BT-02	Genetic Algorithm — Feasible Solution Generation	Pass
Preconditions	The FastAPI backend is running. Gurobi is not required for this scenario.	
Test Steps	1. Select Genetic Algorithm as the optimization method in the request payload. 2. Submit a routing problem instance to the /optimize endpoint. 3. Inspect the returned routes and objective value. 4. Confirm that execution completes without errors or timeouts.	
Expected Result	The heuristic engine returns a feasible routing solution. Objective value is present and routes are non-empty. No unhandled exceptions or 5xx responses are observed.	

BT-03 Custom Dataset Upload — Valid File Acceptance		Pass
Preconditions	A correctly formatted custom routing dataset file is prepared.	
Test Steps	1. Send a POST request to the dataset upload endpoint with the valid file attached. 2. Observe the HTTP status code returned by the backend. 3. Verify that the backend parses the file and acknowledges successful ingestion. 4. Trigger an optimization run using the uploaded dataset to confirm end-to-end usability.	
Expected Result	Backend returns HTTP 200 or 201. The uploaded dataset is correctly parsed and stored. A subsequent optimization request using this dataset executes without errors.	

BT-04 Custom Dataset Upload — Invalid File Rejection		Pass
Preconditions	Incorrectly formatted and incomplete dataset files are prepared for testing.	
Test Steps	1. Submit an incomplete dataset (missing required fields) to the upload endpoint. 2. Submit a file with an unsupported format (e.g., plain text instead of JSON/CSV). 3. For each case, inspect the HTTP status code and response body. 4. Confirm that the backend continues to operate normally after rejecting invalid inputs.	
Expected Result	Backend returns HTTP 400 for each invalid input with a descriptive error message. The application remains stable and subsequent valid requests are processed correctly.	

Frontend Testing

FT-01	Algorithm Selection — Correct Parameter Transmission	Pass
Preconditions	The frontend application is loaded in a supported browser. Backend is running.	
Test Steps	1. Open the algorithm selection menu and choose Gurobi. 2. Inspect the outgoing network request using browser developer tools. 3. Repeat the process selecting Genetic Algorithm. 4. Confirm that the algorithm parameter in each request matches the user selection.	
Expected Result	Each algorithm selection correctly sets the corresponding parameter in the backend request. No mismatches between UI selection and transmitted payload are observed.	

FT-02	Optimization Result Visualization	Pass
Preconditions	A valid optimization request has been submitted and the backend has returned results.	
Test Steps	1. Trigger an optimization run from the frontend interface. 2. Wait for the result visualization page to load. 3. Compare displayed routes, execution time, and objective value against the raw backend response. 4. Scroll through the result view to confirm all data is rendered without truncation.	
Expected Result	The frontend accurately displays routes, execution time, and objective value as returned by the backend. No data is missing, truncated, or incorrectly formatted in the UI.	

FT-03	Algorithm Comparison View — Side-by-Side Metrics Display	Pass
Preconditions	Both Gurobi and Genetic Algorithm results are available for the same dataset.	
Test Steps	1. Navigate to the comparison section of the application. 2. Select the same dataset for both optimization methods. 3. Trigger both optimizations and wait for results. 4. Verify that performance metrics for both algorithms appear side by side without overlap.	
Expected Result	The comparison view correctly renders both algorithm results simultaneously. Objective values and execution times are clearly attributed to their respective methods without visual inconsistencies.	

FT-04	Multilingual Support and Theme Switching — Core Workflow Integrity	Pass
Preconditions	The application is loaded with default language and theme settings.	
Test Steps	1. Switch the application language to an available alternative language. 2. Perform a full optimization workflow (select algorithm, upload dataset, run, view results). 3. Switch the UI theme (e.g., dark/light mode) mid-session. 4. Confirm that the optimization results remain consistent after language and theme changes.	
Expected Result	Language and theme changes do not disrupt the optimization workflow. Results remain accurate and the UI renders correctly under all language and theme combinations tested.	

Semantic Testing

ST-01	Customer Visit Completeness — No Omissions or Duplicates	Pass
Preconditions	A routing dataset with a known number of customer locations is available.	
Test Steps	1. Submit the dataset to both Gurobi and Genetic Algorithm endpoints. 2. Extract the list of visited customer nodes from each generated route set. 3. Compare the visited node list against the full customer list from the input dataset. 4. Check for any customer appearing more than once across all routes.	
Expected Result	Every customer location appears exactly once across the generated routes for both solvers. No customer is omitted and no customer is visited multiple times.	

ST-02	Drone Operational Constraints — Range and Return Compliance	Pass
Preconditions	A dataset containing drone delivery tasks is available. Both solvers are accessible.	
Test Steps	1. Run the optimization with a dataset that includes drone-assignable deliveries. 2. Inspect each drone sub-route in the generated solution. 3. Verify that drone tasks do not exceed the defined operational limits. 4. Confirm that each drone route terminates with a return to the truck node.	
Expected Result	All drone routes comply with operational constraints. No route exceeds defined limits and every drone sub-route ends with a valid return to the associated truck.	

ST-03	Cross-Solver Constraint Equivalence — Gurobi vs Genetic Algorithm	Pass
Preconditions	The same routing dataset is available for submission to both optimization methods.	
Test Steps	1. Submit the dataset to the Gurobi optimizer and record the generated solution. 2. Submit the same dataset to the Genetic Algorithm and record its solution. 3. Verify that both solutions satisfy the same set of routing constraints. 4. Compare objective values and note the difference; confirm neither solution is logically invalid.	
Expected Result	Both solvers produce solutions that satisfy all routing constraints. Objective values differ as expected between exact and heuristic methods, but neither solution contains logical violations such as unvisited customers, constraint breaches, or invalid drone assignments.	

ST-04 Solution Validity Under Multiple Datasets — Consistency Check		Pass
Preconditions	At least three distinct routing datasets of varying size and complexity are prepared.	
Test Steps	1. Submit each dataset individually to both optimization solvers. 2. For every generated solution, verify customer visit completeness. 3. For every generated solution, verify drone constraint compliance. 4. Record whether each solution passes all semantic checks.	
Expected Result	All generated solutions across all tested datasets pass semantic validation. Customer completeness and drone constraint compliance are maintained consistently regardless of dataset size or complexity.	
